LAB EXPERIMENT
ITA0506-COMPUTER VISION

PAVENDHAN A -192321109

EXPNO:25

Aim

To sharpen an image using a High-Boost Masking technique.

Principle

High-boost filtering is an extension of unsharp masking. It increases the contribution of the original image while adding the high-frequency details.

The formula is:

where:

- = Original image
- = Blurred image
- = High-boost constant
- = Sharpened image
-

When A = 1, it becomes standard Laplacian/unsharp sharpening.

High-Boost Masks

As shown in your image, two masks can be used.

4-neighbor mask:

0 -1 0

-1 A+4 -1

0 -1 0

8-neighbor mask:

-1 -1 -1

-1 A+8 -1

-1 -1 -1

Here, we will use A = 2.

Algorithm

1. Read the input image.
2. Convert the image to grayscale.

3. Select the high-boost value .
4. Create the high-boost mask.
5. Apply the mask to the image using convolution.
6. Limit pixel values between 0 and 255.
7. Display the original and sharpened images.

CODE:

EXP-25.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure BEN 10.png is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Set high-boost constant
A = 2

# Step 4: Create High-Boost Mask
# 4-neighbor mask
mask = np.array([
    [0, -1,  0],
    [-1, A + 4, -1],
    [0, -1,  0]
], dtype=np.float32)

# Step 5: Apply high-boost mask
sharpened = cv2.filter2D(gray, cv2.CV_32F, mask)

# Step 6: Convert result to valid pixel range
sharpened = np.clip(sharpened, 0, 255)

# Convert to unsigned 8-bit
sharpened = sharpened.astype(np.uint8)

# Step 7: Display images
cv2.imshow("Original Image", gray)
cv2.imshow("High-Boost Sharpened Image", sharpened)

# Step 8: Wait for key press
cv2.waitKey(0)
```

Ln: 10 Col: 31

OUTPUT :

Result

The image is successfully sharpened using a High-Boost Mask. The high-boost mask enhances edges and fine details while retaining the main information of the original image.